VIA

Illustrated Assembly and Demo Handbook

OnePlus Pad OPD2203 · Hardware R14.1 · Applications 1.0.5
Expanded documentation edition · 8 November 2026



A practical companion to the engineering rationale: part selection, printed joints, hidden wiring, tablet service and the local phone/tablet demonstration. Actual-mesh illustrations communicate geometry; physical qualification remains a separate step.

github.com/vluncasu/via-oneplus-pad-terminal

1. VIA Engineering Documentation

This handbook connects the OnePlus Pad OPD2203 enclosure to its printable components, assembly interfaces and offline Android demonstration. Hardware revision remains R14.1; this documentation expansion does not change the STL geometry or APKs.

Download the illustrated handbook PDF for the offline assembly/demo reference. The separate engineering report develops the design rationale.

Expanded offline downloads · Edition changes

Choose your task

Task	Read first	Use alongside it
Understand the object	Design rationale	Product views
Identify every part and quantity	Part catalog	Color-separated files
Decide what to print first	Manufacturing workflow	Printing settings
Understand joints and tablet retention	Interfaces and retention	Engineering views
Assemble without losing access	Illustrated assembly	Fastener map
Install LED strips and internal leads	LED and wiring	Dimensions and clearances
Run the ten-minute demonstration	Demo runbook	Software architecture
Diagnose a problem	Troubleshooting	Acceptance and test plan
Audit a design claim	Requirements traceability	Verification record
Edit or reproduce the design	Direct editing guide	CAD formats and validation

What each evidence label means

Modeled means the feature exists in the supplied shape. Digitally checked refers to a named report or build/test result. Trial procedure is an instruction for a future physical check. Reference envelope reserves space; it is neither a real component model nor a promise that a purchased component fits. Keep these distinctions when presenting the project.

Required color batches are 25 white PETG pieces, 4 black PETG pieces and 1 natural PETG film. These are quantity totals, not numbers of separate STL files.

The normal Blender file opens as an assembled terminal. Isolated, cover-omitted and sectioned figures are supplementary inspection views. No illustrated cutaway is an STL modification. Parts marked R01–R05 are reference envelopes and are not printed.

Version and scope

- Application source / official demo APKs: 1.0.5.

- Mechanical baseline: R14; O05 optical film amended to 0.20 mm in R14.1.

- Required print set: 17 unique files, 30 printed pieces; 15 optional probes.

- Runtime: local tablet and phone, with a QR request and BLE response; no external backend.

- Actual printing, tablet fit, load capacity, temperatures and radio/camera behavior require physical qualification.

Start with the low-material probes and an unpowered dry build. Keep the same part IDs in your slicer project, build notes and test photographs so a failed interface can be traced to an actual file.

2. Printable Part Catalog

Parts, quantities and filament groups for the R14.1 enclosure. Bounds are X × Y × Z in the supplied STL print orientation, not assembly dimensions. Numbers are rounded to two decimals. Quantity is the total copies of the one file; reference objects R01–R05 are excluded.

Required color group	Unique files	Printed pieces
white PETG	12	25
black PETG	4	4
natural PETG	1	1

Required components

IDs	Part / file	Copies	Filament	Supplied STL bounds (mm)
B01	monoblock base integrated light belt	1	white PETG	190.21 × 190.21 × 230.00
B03	bottom service cover	1	white PETG	157.26 × 157.26 × 209.40
B07	integrated s stand and power supply cover	1	white PETG	240.00 × 230.00 × 197.00
P01	recessed dummy port cartridge and cable funnel	1	black PETG	180.00 × 19.99 × 14.00
H01	structural front frame integrated led channel	1	white PETG	242.75 × 242.75 × 224.00
H04	lower structural rear cover	1	white PETG	221.61 × 221.61 × 187.55
O01	open optical carrier	1	black PETG	26.00 × 194.00 × 29.80
O03	optical carrier usb service cover	1	black PETG	34.60 × 202.60 × 7.00
V21, V22, V23, V24	head mount screw D12 P2p5 L12	4	white PETG	18.71 × 18.72 × 16.00
V31, V32, V33, V34	spherical slotted screw foot D8	4	white PETG	23.23 × 23.23 × 20.00
V41, V42, V43, V44	base cover screw D8 P2p5 L10	4	white PETG	12.48 × 12.48 × 14.00
V51, V52, V53, V54	rear cover screw D8 P2p5 L14	4	white PETG	12.48 × 12.48 × 18.00
V61, V62	usb cover screw D8 P2p5 L16p3	2	white PETG	12.48 × 12.48 × 20.30
H05	sliding top tablet retention bar	1	white PETG	195.16 × 195.16 × 8.30
O05	captive optical film 0p20mm	1	natural PETG	25.60 × 193.60 × 0.20
H06	upper rear service cover integrated locking pins	1	white PETG	236.58 × 236.58 × 34.20
O06	flat back dummy sensor face	1	black PETG	26.00 × 194.00 × 3.00

Optional low-material probes

IDs	Part / file	Copies	Filament	Supplied STL bounds (mm)
Q12	usb plug pull through gauge 16x10x30	1	white PETG	16.00 × 10.00 × 30.00
Q01	D12 thread test screw	1	white PETG	18.71 × 18.72 × 16.00
Q02	D12 nut clearance 0p25	1	white PETG	19.99 × 20.00 × 10.00
Q03	D12 nut clearance 0p35	1	white PETG	19.99 × 20.00 × 10.00
Q04	D12 nut clearance 0p45	1	white PETG	19.99 × 20.00 × 10.00
Q05	D8 thread test screw	1	white PETG	12.48 × 12.48 × 14.00
Q06	D8 nut clearance 0p30	1	white PETG	13.99 × 14.00 × 8.00
Q07	tablet corner fit coupon	1	white PETG	30.00 × 30.00 × 9.00
Q11	front camera clearance coupon	1	white PETG	40.00 × 17.89 × 8.30
Q08	light transmission wall 0p6	1	white PETG	25.00 × 15.00 × 0.60
Q09	light transmission wall 0p8	1	white PETG	25.00 × 15.00 × 0.80
Q10	light transmission wall 1p0	1	white PETG	25.00 × 15.00 × 1.00
Q13	tablet width gauge 259p03	1	white PETG	200.13 × 200.13 × 11.00
Q14	tablet height gauge 190p41	1	white PETG	151.61 × 151.61 × 11.00
Q15	led channel corner coupon 8p5	1	white PETG	29.00 × 24.00 × 14.00

Alternatives and reference envelopes

O05 0.40 mm is an optional replacement for the 0.20 mm default, not an extra required piece. R01 tablet, R02 supply, R03 USB plug and R04/R05 LED volumes are inspection references, not printable hardware.

White PETG: structural pieces and printed fasteners. Black PETG: P01, O01, O03 and O06. Natural PETG: O05. The present rigid feet are not TPU pads.

Reading the file families

- B: base / stand and service enclosure.

- H: structural head and tablet retention.

- O: optical cassette, service cover, sensor face and film.

- P: one-piece dummy port cartridge.

- V: printed screw / foot instances.

- Q: optional fit and process probes.

- R: reference envelopes, never a print quantity.

Use fasteners and service for the receiving interface of each V family, manufacturing workflow before large prints, and assembly for the build order.

This catalog is a file-selection aid. Its bounds and closed-mesh results do not qualify strength, fit, supports or optical performance.

3. Dimensions, Datums and Clearance Budget



Figure 14. Assembled front-facing orthographic view. No dimensions should be measured from perspective product figures.

All dimensions are millimetres. Values below distinguish manufacturer body dimensions, nominal CAD envelopes and unvalidated allowances. A CAD clearance is not automatically a usable manufacturing tolerance.

Feature	Value / interpretation	Source and verification needed
OnePlus Pad OPD2203 body	258.03 × 189.41 × 6.54	Manufacturer nominal; measure the actual tablet and protrusions
H01 head envelope	About 310 × 224 × 33.3 in head coordinates	R14 geometry, not the earlier 290 mm concept
Base plan envelope	240 × 230	Assembled CAD footprint; differs from rotated export bounds
Tablet opening gauges	259.03 width / 190.41 height	Nominal +1.00 mm overall relative to body; not a complete fillet/button fit specification
USB body gauge Q12	16 × 10 × 30	Provisional reference, not a universal USB-C plug size
Stand duct	About 30 wide × 24 central height	Shaped route; local transitions and connector rotation matter
Supply reference R02	92 × 64.02 × 39.02	Body envelope only; terminals, access and cooling remain to be verified
LED strip width	8.0	User-provided width; voltage, package height and bend limit still unspecified
LED reserved width	8.5	0.5 total nominal allowance, not 0.5 per side

Feature	Value / interpretation	Source and verification needed
Visible light belt	About 3	Optical appearance target, not strip width
O05 current film	25.6 × 193.6 × 0.20	R14.1 single-layer amendment

Datums for checking an assembly

Use the desk plane, the base exterior, the head plane and the tablet front glass as distinct references. The provided STL bounding box is measured in its exported print orientation. The native CAD and assembly meshes use assembly coordinates. Do not derive a joint's assembled angle by comparing two rotated STL boxes.

The head display plane is nominally 60° to the desk. The nominal P1S cube is 256 × 256 × 256. A 310 × 224 × 33.3 head rotated edge-on at 45° has an approximate plan projection of (310 + 33.3) / sqrt(2) = 242.75 in both X and Y, with 224 height. Brims, supports, exclusion zones and adhesion space are additional; a bounding-box pass does not establish a usable print job.

Fit budget

Account separately for extrusion and shrinkage, first-layer expansion, paint thickness, guide clearance and actual component variation. A 1.00 mm total opening allowance gives about 0.50 mm per side only for a centered ideal body in a straight section; corner radii, glass overlap and buttons can govern first contact instead.

Use probe results to decide local adjustments. Do not scale a whole part just to gain a few tenths at one opening: doing so also changes coarse threads, LED seat width and mating datums. Keep optical film thickness changes separate from its outline; the current amendment preserves the film outline and front datum.

Connector passage

Prove passage with the complete actual connector, not only the flexible cable. Pull the unpowered connector through the stand, head junction and optical-service opening, preserving a safe cable bend. Then add the LED leads and repeat with all required wires present. The path must remain serviceable after covers are installed. A cable should not become the spacer that prevents a structural face from seating.

Record the smallest successful gap and the actual plug model in your build notes. There is no universal “standard body plus reserve” dimension for every molded USB-C plug.

4. Manufacturing Workflow and Print-Preparation Gates

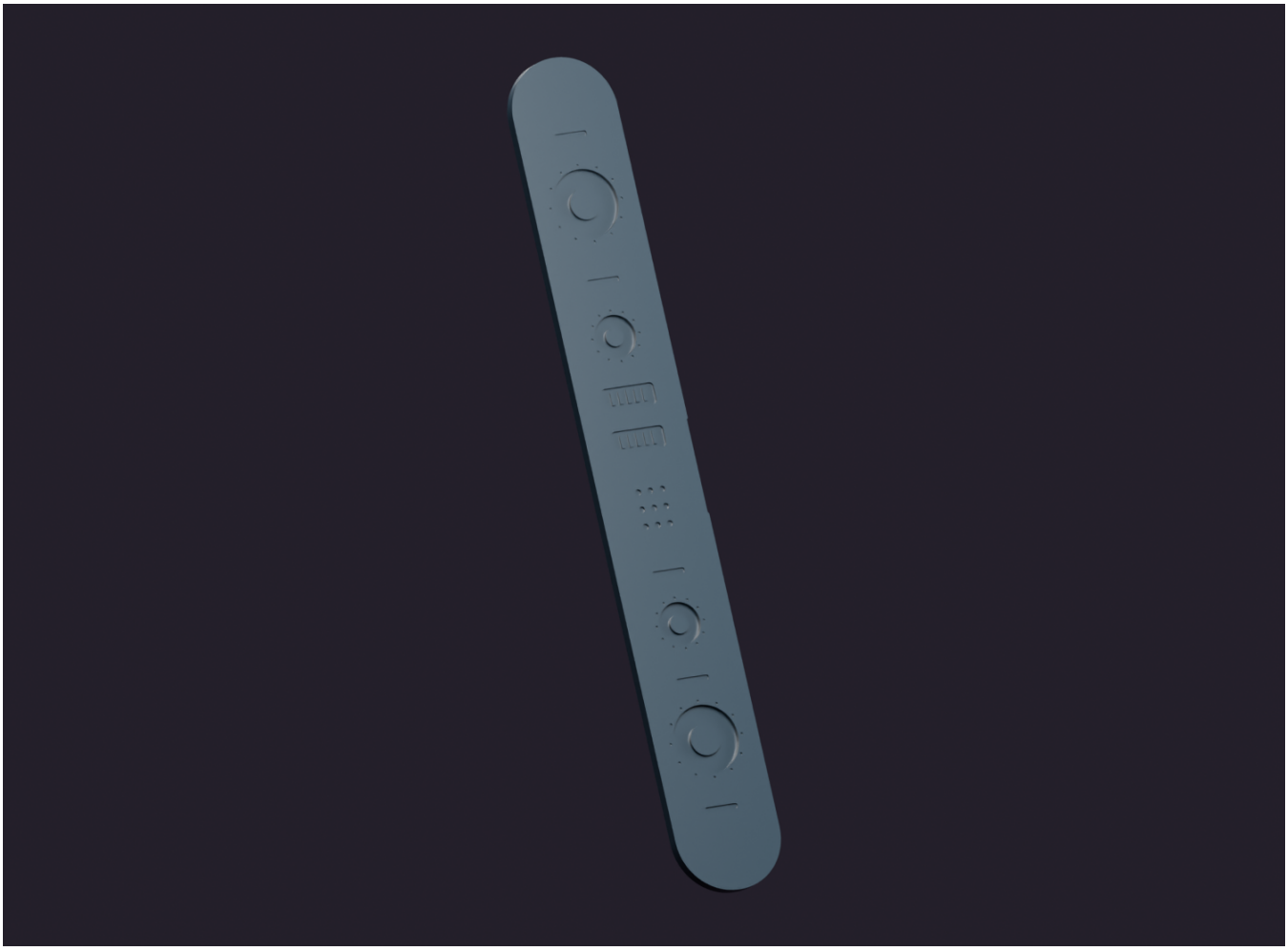


Figure 11. O06 black printed face with sensor-like relief; no real stereo camera or additional sensors are implied.

Use this sequence to avoid committing the largest prints before the interfaces are understood. It is a trial workflow for the published meshes, not a prevalidated printer profile. The catalog lists the actual supplied STL bounds, which can differ from assembled dimensions because parts are rotated for printing.

1. Establish the reference components

Record the actual tablet width, height, body thickness, corner shape, rear camera protrusion, button positions and front-camera field of view. Record the complete USB-C plug body including strain relief and any right-angle projection. Measure the supply body and the additional space needed for terminals, connectors and cooling. A 92 × 64.02 × 39.02 mm CAD block is not a mounting drawing for every 12 V supply.

Keep an unmodified copy of the release files. Import in millimetres and check a known dimension before changing orientation or scale. A near-fit should be corrected at the responsible interface, not by uniformly scaling the terminal and changing every thread and seat.

2. Print the smallest useful evidence

Question	First probe	What constitutes a useful result
Will the tablet corner and thickness seat?	Q07	Real corner seats without glass pressure; rear protrusions remain accounted for
Is the whole opening large enough?	Q13 and Q14	Actual body clears both gauges; they do not establish all corner/button clearances
Is the front camera unobstructed?	Q11	Preview has no edge obstruction in the intended mounting position
Can the cable body pass through?	Q12 plus the actual unpowered cable	Both pass the complete route; a straight gauge alone does not establish a bent route
Which thread clearance works?	Q01–Q06	Starts by hand, seats repeatedly, does not strip or bind
Which wall transmits acceptable light?	Q08–Q10	Compare with the exact strip and filament under the intended room lighting
Does the strip corner fit?	Q15	Width, package height, bend direction and feed joint all fit

These coupons are inherited checks. H05/H06 guide and pin engagement still need a local joint sample or a careful first complete dry build. A successful Q12 print is not proof that every USB-C cable is compatible.

3. Prepare each structural job

Inspect the contact face on the plate and the highest bridge in the slicer. H01/H04 use an edge-on diagonal strategy; B01/B03 supplied files also have deliberate rotations. Keep the proposed orientation until supports, cooling and bed exclusions have been reviewed. B07 is a single-piece stand plus upper power-volume cover, so a print failure near its root affects the whole part.

Use the printing guide for trial layer heights and walls. Start with the filament maker's profile, calibrate flow and dimensional behavior, and inspect the first layers before leaving a long job. Add supports only after checking whether they can be reached and removed. Do not seal support inside the stand duct, LED grooves or optical carrier.

4. Distinguish CAD voids from slicer infill

The stand duct, supply cavity, LED seating regions and service openings are modeled empty regions. They are not low-density infill regions. In the layer preview they must remain empty, except for explicitly generated supports that can be removed. If infill appears in the duct, inspect the imported mesh interpretation and any repair operation before continuing.

The structural shell surrounding those voids receives perimeters and infill. For a first trial, use the guide's 5–6 walls and 15–25% gyroid, then review local modifiers at the stand root, head interface and thread seats. Use continuous material around a threaded boss and its supporting web. Avoid a tiny dense cylinder disconnected from the surrounding wall; density alone does not make a load path.

Region	Preserve	Inspect in layer preview
B07 stand root	Braces, radiused transitions and duct walls	Continuous layers into the upper cover; no support trapped in the duct
H01 stand interface	Thread seats and their connecting material	More than isolated towers; no LED feed in the bearing region
LED channels	Seating and optical rims	Strip clearance, corner transitions and accessible feed point
Fine vents	Thin concealment geometry and open flow paths	Features are resolved by the 0.4 mm nozzle rather than silently omitted

Region	Preserve	Inspect in layer preview
O01/O06 optical parts	Flat printing surfaces and captive shoulder relationship	Open carrier; details upward on O06; no large roof across its cavity

5. Print optical and repeated parts

Print O06 black with its flat back down. Keep detail surfaces upward; begin at 0.08–0.12 mm layers and check whether narrow features survive the preview. O01 is the open carrier, not a closed sensor box requiring a roof full of supports. P01 is one black cartridge containing all dummy ports.

Print O05 natural PETG at 0.20 mm first layer and 0.20 mm total height. The preview must contain exactly one layer. Do not add a raft or optical-face supports. Keep the 0.40 mm alternative separate; it is not a second required film. Printed PETG film will not reproduce the optical quality of glass.

Only print all repeated screws after a thread probe is usable. Filename quantity is the count of copies of that geometry; for example, four copies of the V21-family file provide V21–V24 in total.

6. Finish, inspect and record

Remove burrs without rounding away guides or thread starts. Trial assembly before filler, sanding or paint. Mask threads, locating faces, luminous zones and the film seat. Record material, nozzle, layer height, wall count, local modifiers, support strategy and orientation for each successful part. Slicer estimates are the source for actual filament use and print duration; CAD solid volume is not a spool estimate.

Proceed to the assembly sequence, then sign off the acceptance checks. Keep failed probe settings in your notes: they explain which clearance and material combination was rejected.

5. Mechanical Interfaces, Retention and Load Paths

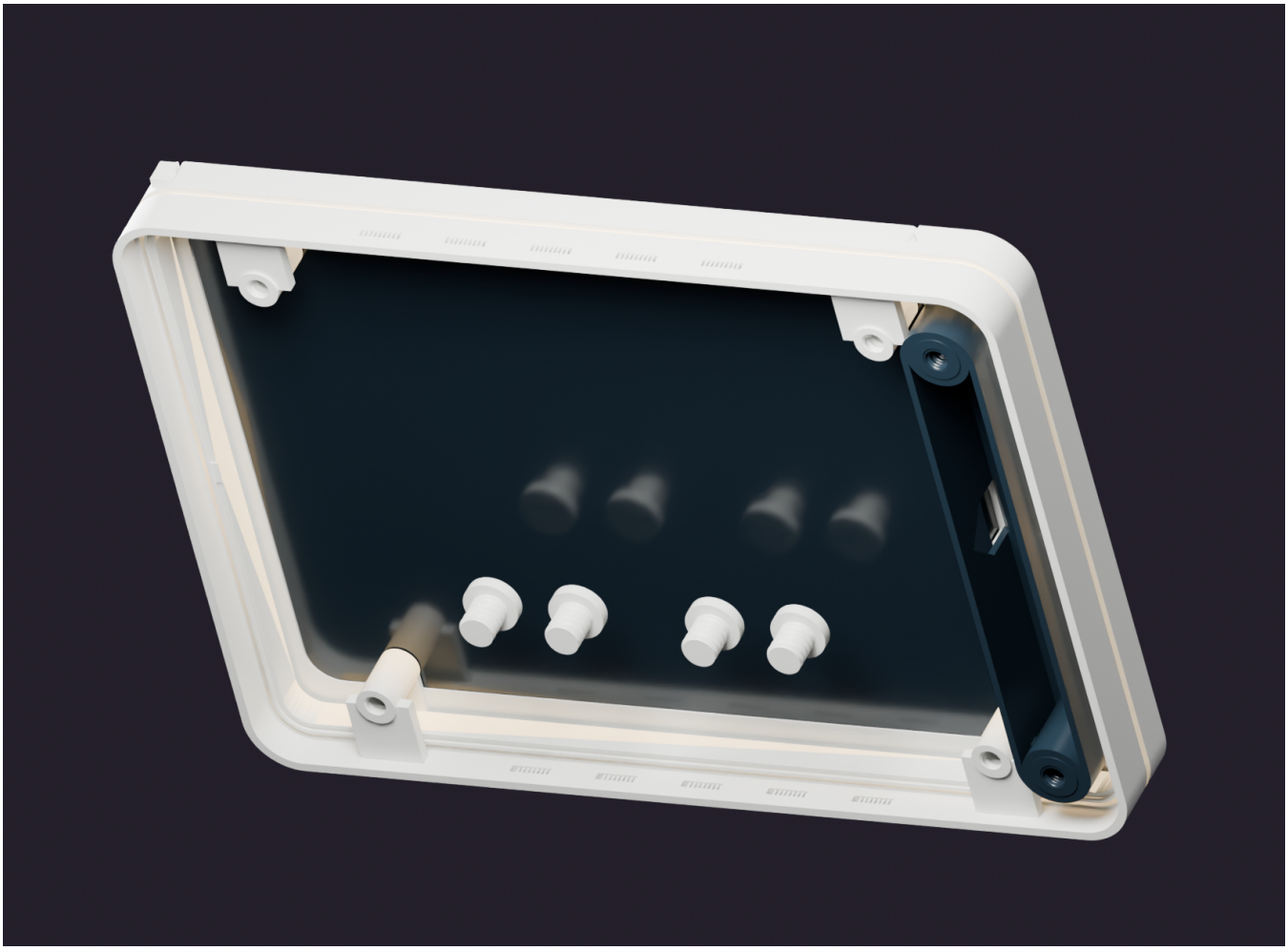
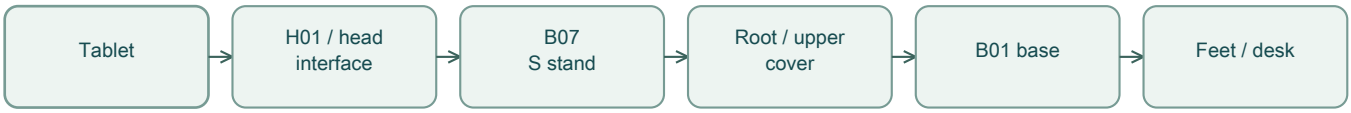


Figure 20. H01, nominal R01 tablet envelope, optical parts, H05 and head screws remain. H04/H06 omitted for seating inspection.

The assembly has three different jobs: carry the tablet's loads, retain service parts, and hide decorative details. A locating surface positions a part; a threaded joint retains it; neither should rely on trapped wiring. The engineering gallery shows these relationships with selected covers omitted.

Structural chain



Intended interface relationships; the wiring view is a functional topology, not an electrical schematic.

This is the intended load path, not a structural calculation establishing capacity. The power-supply reference is not a brace and must not support the stand by accidental contact. The root reinforcement belongs to B07. It should transfer force into the broad cover while preserving both the explicit duct and the intended electronics space.

Base and power-volume cover

B01 is the monoblock outer base. B07 integrates the upper volume and the one-piece S stand. B03 closes the service side underneath. The outer interfaces should seat before their fasteners are tightened. The top finish is not a place to add compensating screw holes. Feed wires and the cartridge are installed while the underside remains accessible.

The underside view with the orange block depicts R02, a provisional supply envelope. It makes the conflict between reinforcement, component space and access visible. It does not represent terminal clearance, mains isolation or thermal certification of an actual supply. Verify these with the selected low-voltage power system and its manufacturer's instructions.

Head-to-stand connection

The head is attached by the V21–V24 family of coarse D12 printed screws at the matched interface. Check that all intended bearing faces contact evenly. If the joint rocks while a screw is loose, inspect burrs or a geometric mismatch; tightening harder is not a corrective alignment method. Keep the internal cable passage open across this connection and pull the cable through before final closure.

A visible junction line is a service interface. It should remain narrow and consistent, without a projecting bracket being mistaken for extra stand width. This publication depicts the actual R14 interface; it does not replace its geometry with a seamless artistic blend in the renders.

Tablet restraint and upper-rear service

H01 surrounds the original tablet face, while rear covers/stops control rearward movement. The actual tablet's own front glass remains exposed. The frontal camera opening belongs to the actual camera position, not to the decorative side sensor bar.

H05 is a sliding top retention bar. Its open guides constrain it through the depth direction while permitting the intended upward removal path once unlocked. H06 is the upper rear service cover. Its integrated straight pins enter H05 receivers and block the bar's lifting path when the cover is fully seated. The rear screws retain H06. These pins are not a flexible snap mechanism and there is no separate tiny pin to install.

State	Parts present	Intended outcome
Closed	H05 guided, H06 seated and rear screws installed	Bar cannot simply lift out; upper stops retain the tablet
Service cover removed	H06 withdrawn, tablet supported	Integrated pins clear H05; bar can follow its guide path
Extraction	H05 removed, USB-C unplugged, tablet supported	Tablet leaves through the upper rear service opening

Do not test removal by pulling on the tablet while the service cover is fitted. First support the assembly, disconnect USB-C through O03, undo the two upper rear screws, withdraw H06, then lift H05 in its screen plane. The archived sweep report is only a nominal geometric sweep: real camera protrusions, buttons, cable stiffness and printed deformation are excluded.

Captive optical assembly

O01 is the black open carrier. O06 is the separate flat-backed decorative face. O05 is the 0.20 mm film, captured against the shoulder integrated into H01. O03 closes the USB service region using V61/V62. The structural shoulder and assembled carrier relationship are intended to prevent a forward fall; the printed film is not an adhesive structural retainer.

Test capture from both sides during the dry build. With the tablet absent, gently check whether the film, face or carrier can leave its intended seat. The thinner R14.1 film has a different flexibility from the original 0.40 mm film, so clearance and capture must be verified again. If it falls forward, the assembly is not accepted merely because the render appears flush.

The side cassette hides the USB-C connector and cable, not the whole charger. Its dummy lenses, recesses and sensor-like relief are decorative. The functional face/smile/palm camera is the tablet's front camera.

Recessed port cartridge

P01 is one black printed part with the dummy connector internals and real cable funnel. It is inserted through the service side into the base's recessed opening. Check recess depth, locating contact and closure of the bottom assembly before judging retention. Do not depend on a friction fit whose behavior has not been measured. If the cartridge can move outward after closure, record the actual contact gap and revise the responsible retaining geometry.

Real electrical connectors are not formed by the printed dummy features. Their internal tongues/cavities create depth and visual detail. Avoid testing them with a real plug; thin printed features may break or trap it.

See fasteners and service for screw families and acceptance checks for practical verification.

6. Assembly and Service Instructions

Use the BOM and part IDs. Interior markings are intended for identification; verify that any printed marking is readable before finishing. Reference envelopes R01–R05 are not printable components.

Assembly map

Identify parts → qualify probes → build base/stand → route leads → mount head → fit optical parts/tablet → close service covers → acceptance checks. Keep the underside open until leads and cartridge are installed.

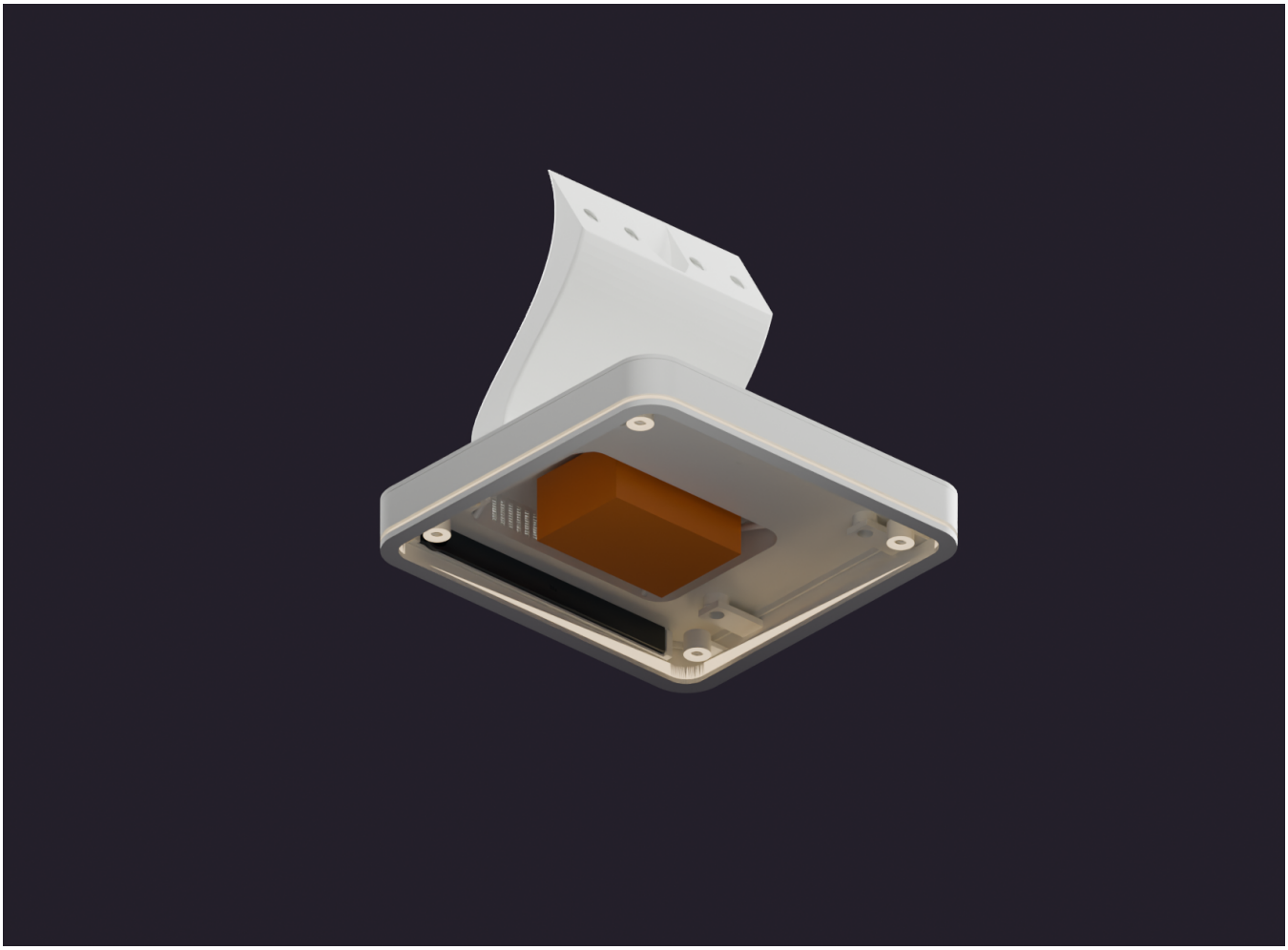


Figure 22. B01/B07/P01 with orange R02 body envelope; bottom service cover omitted. R02 is not a detailed supply or a clearance qualification.

Figure 22 omits B03; orange is the provisional R02 body envelope, not a selected electrical installation.

Dry assembly order

1. Print the relevant fit, connector, thread and LED coupons. Confirm the actual OPD2203 dimensions, front-camera position, USB-C plug body, rear protrusion and chosen supply.

2. Inspect and clean B01/B03/B07; check that the stand duct is continuous and contains no support remnants or infill. Pull the Q12 gauge, then the actual unpowered connector/cable, through the complete route.

3. Insert P01 from the service side into the recessed base opening. The port face should remain within the base's exterior envelope; do not force a protruding cartridge or insert real plugs into the dummy cavities.

4. Trial the chosen power module against its actual mounting/clearance needs. Keep the S-root braces intact. Install low-voltage wiring and strain relief so the service cover cannot pinch the cable. The reference supply is only an envelope, not a validated mains installation.

5. Seat the B07 integrated upper cover/stand in B01, checking the perimeter alignment and retained cable route. Use the designed underside-access fastening arrangement; do not drill new holes through visible top faces to compensate for poor fit. Printed threads require calibrated coupons and gentle hand tightening.

6. Attach the head at its matched stand interface using V21–V24, retaining internal wire routing. Confirm the joint bears on its intended structural faces rather than on a cable or cosmetic edge.

7. Install and test the LED strips as described in LED and wiring. Do this before final closure; do not route the corner feed through a screw seat.

8. Fit the side cassette from the service side: O05 film against the integrated H01 shoulder, O06 decorative face, O01 open carrier, then O03 USB service cover with V61/V62. Verify captive retention from both front and back. The thinner 0.20 mm film needs a fit test; it is not adhesively bonded over the tablet screen.

9. Support the tablet and slide its body into the head. Keep the real front camera clear. Connect USB-C through the optical/service region; the entire charger is not housed in the sensor bar.

10. Install H04, then H05 in its top guides. Fit H06 fully so its integrated pins enter the bar's blind receivers and the tablet stops seat properly. Fasten the rear covers with V51–V54 as appropriate; do not use a screw to pull a misaligned cover into place.

11. Close B03 using V41–V44. Fit the four V31–V34 spherical screw feet, keeping their driver slots accessible. Check desk contact and sliding stability.

12. Verify charging, camera view, lighting and the software demonstration while monitoring temperature. Finish and paint only after successful repeated dry assembly.

Inspect the tablet and retention interfaces

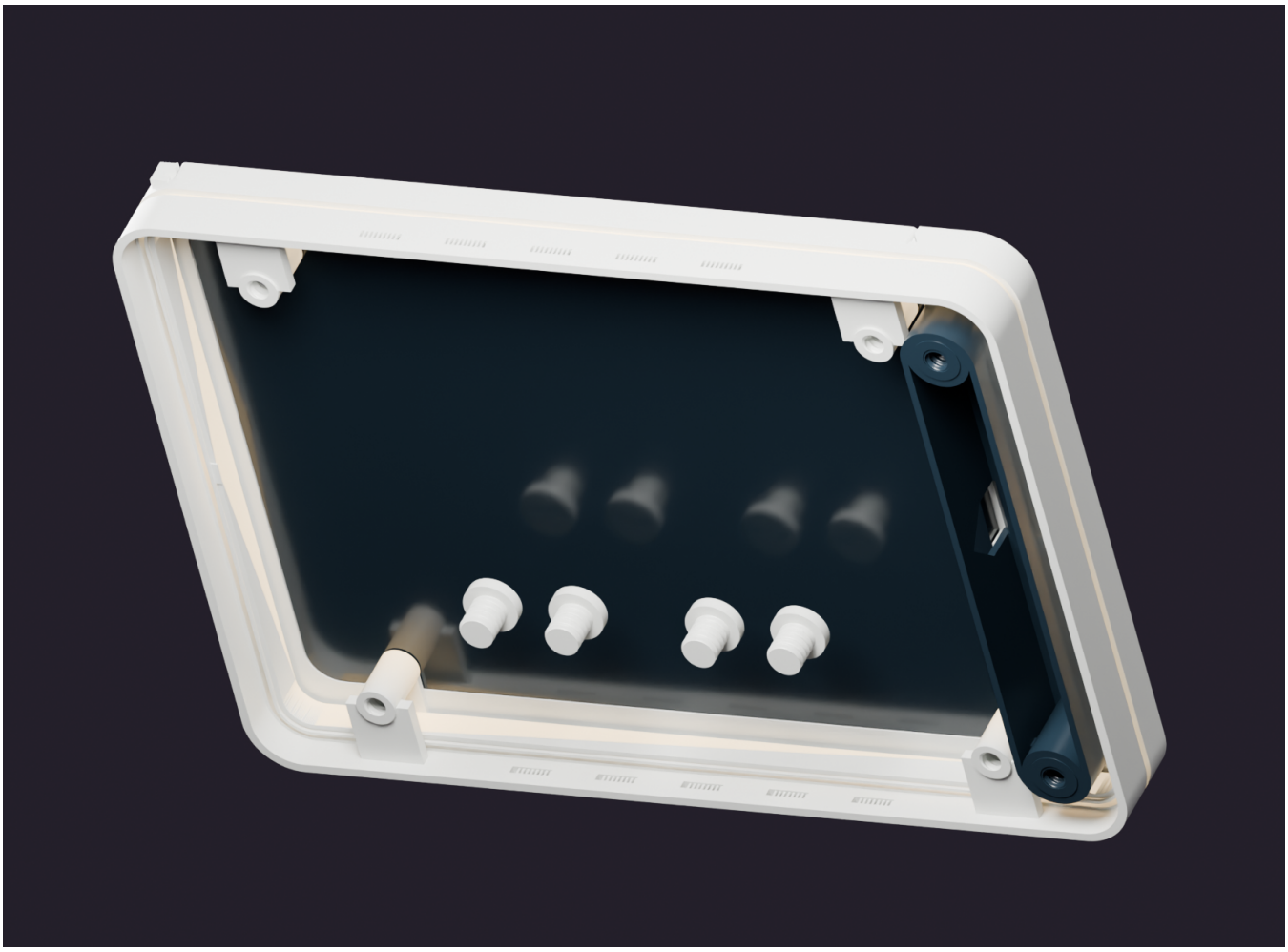


Figure 20. H01, nominal R01 tablet envelope, optical parts, H05 and head screws remain. H04/H06 omitted for seating inspection.

Figure 20 retains the nominal tablet body and H05, omitting rear covers for inspection.



Figure 21. H01/H04/H05/H06 with rear screws in assembly coordinates. Tablet and optical cassette omitted.

Figure 21 removes the tablet and optical cassette. Use the interface guide to distinguish guides, locking pins and screw retention.

Remove the tablet through the upper rear opening

- Support the head on a padded surface and support the tablet throughout removal.

- Open O03 and disconnect USB-C from the tablet.

- Undo the two upper rear screws V52 and V54.

- Withdraw H06 rearward, freeing its integrated pins from H05 and retracting the upper tablet stops.

- Slide H05 upward in the screen plane and out of its guides. Do not pry it toward the glass.

- Support the tablet edge through the upper rear opening and lift the tablet out.

Reassembly reverses the sequence. The guide pins lock the bar's lift path; they are neither springs nor substitute screws. The archived service sweep excludes real cable bending, buttons and camera protrusions.

Acceptance checks

The port cartridge must not project from the base; neither the optical face nor the film may fall forward; the stand root must not rock; screws must seat without stripping; no wire may cross the tablet extraction path; and service covers must close without loading the glass. Inspect again after a warm closed-enclosure test. No tightening torque or structural capacity has been experimentally established.

Screw and inspection references

Use the fastener map to select the receiving interface and correct length. The troubleshooting guide covers proud screws, loose cartridges and a blocked sliding bar. Complete the acceptance record before the powered demonstration.

7. Printed Fasteners and Service Map

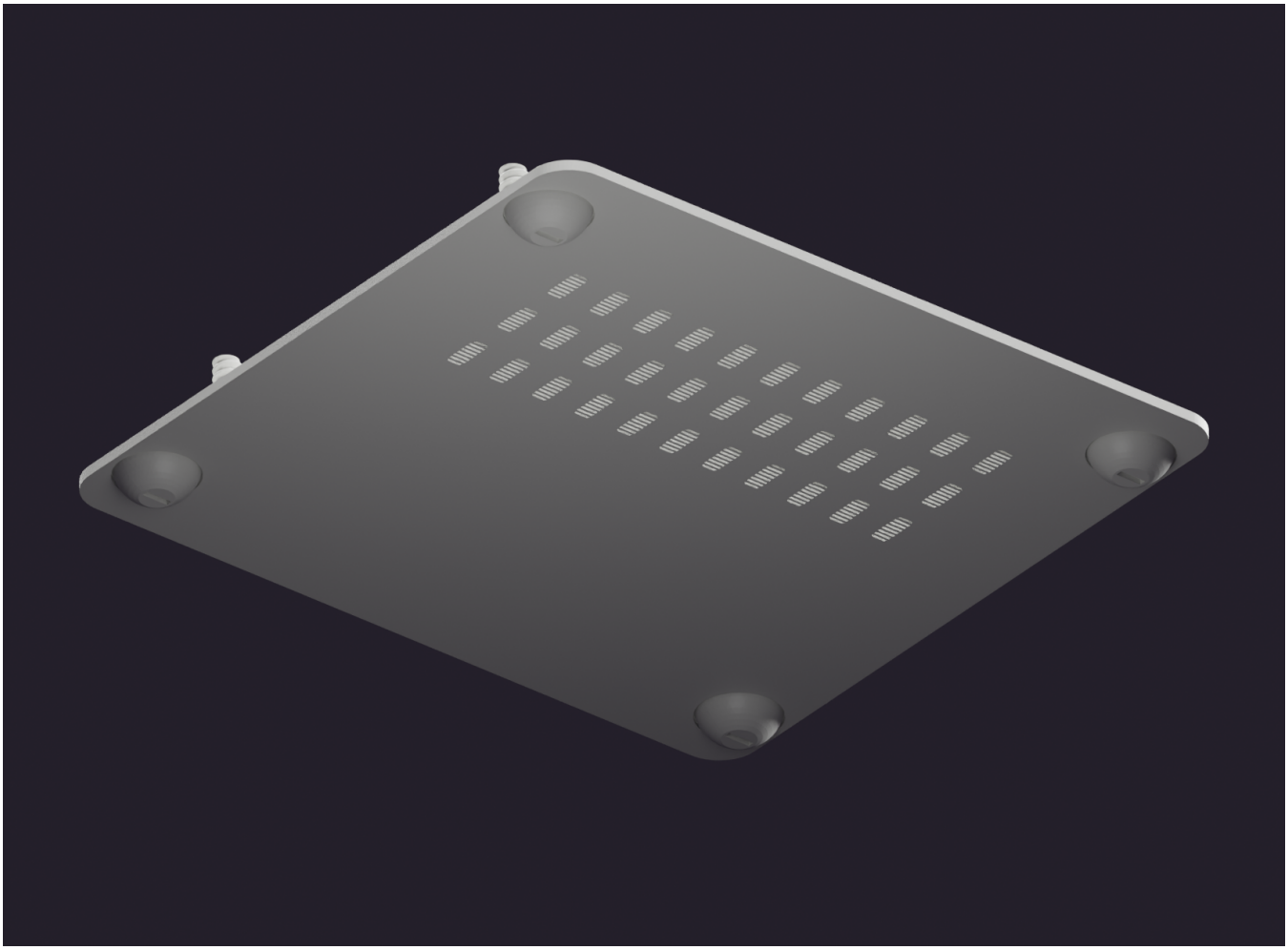


Figure 24. B03 with V41–V44 and V31–V34 in their assembled positions; base and upper cover omitted.

The current assembly uses printed screw families rather than purchased bolts or heat-set inserts. These coarse printed threads are not standard ISO metric hardware. Do not substitute a similarly named metal M8 or M12 bolt without redesigning its mating geometry.

Screw families

IDs / copies	Function	Nominal file designation	Service access
V21–V24 / 4	Head-to-stand structural attachment	D12, pitch 2.5, L12	At the head interface with service access
V31–V34 / 4	Spherical slotted feet	D8 screw foot	Underside; slot must remain accessible
V41–V44 / 4	Bottom service cover	D8, pitch 2.5, L10	Upward from the underside
V51–V54 / 4	Rear head covers / stops	D8, pitch 2.5, L14	Rear face; upper pair V52/V54 releases H06
V61–V62 / 2	USB optical service cover	D8, pitch 2.5, L16.3	O03 service region

File designation is a nominal CAD identifier, not a measured fit or a tightening-torque rating. The full part filenames and quantities are in the catalog.

Fit before force

Print Q01–Q06 using the intended filament, orientation and layer height. Clean the thread start, start square by hand and test repeated insertion into the coupon. Choose a working clearance before making the full screw batch. Use a driver that fits the printed slot and apply gentle hand force; do not use a powered driver or use screw torque to close a misaligned housing.

Check that the screw actually engages the receiving thread, rather than just resting in a cosmetic recess. A rear screw should seat at its intended exterior datum. If it remains proud, inspect debris, wrong family, bottoming length and cover alignment. Grinding the cap flush without checking engagement can remove useful material while leaving the joint loose.

Symptom	Inspect before changing dimensions
Thread will not start	Wrong family, tilted start, elephant foot, seam blob, calibrated clearance
Head seats but joint moves	Mating faces, actual thread engagement, premature bottoming, stripped receiving wall
Cover distorts while tightening	Cable beneath the cover, support remnant, wrong screw length or interface mismatch
Foot loosens repeatedly	Fit, cyclic rotation and desk friction; rigid PETG feet are not anti-slip TPU
Thread cracks its seat	Layer direction, wall continuity, local web geometry and excessive tightening

Service without dismantling everything

Task	Open first	Preserve
Unplug tablet charging	O03, V61/V62	Optical film and tablet front glass
Remove tablet	O03, then H06 / V52/V54, then H05	Support tablet while the top restraint is released
Inspect low-voltage distribution	Bottom cover B03 / V41–V44	Stand-bearing interface and root braces
Inspect optical capture	O03 and the service side of O01, with tablet protected	H01 shoulder and film outline
Inspect port cartridge	Bottom service access	Exterior base wall and cartridge recess

Keep a labeled tray with one compartment per screw family. Reusing the wrong D8 length can bottom out even when the diameter appears correct. Interior part IDs and filename quantities are the primary assembly references. Do not add visible labels or permanent filler over service seams.

No fatigue life, pull-out capacity or torque specification has been established. Inspect printed threads after warm operation and after repeated service; replace damaged pieces rather than forcing them.

8. Integrated LED Channels, Optical Belts and Internal Wiring

Dimensions are different functions

Feature	Nominal design intent
Strip PCB width	8 mm
Reserved seating width / reference	8.5 mm, including 0.5 mm total nominal allowance
Visible luminous belt	Approximately 3 mm
O05 optical film	0.20 mm default; unrelated to the tablet display glass
Stand cable route	Approximately 30 mm wide × 24 mm central height, with a shaped roof
Connector test body	16 × 10 × 30 mm provisional body, Q12

H01 and B01 contain the head/base channel and light-belt geometry. Their integrated shielding/rim geometry is part of those same printed structures. There are no separate L1/L2 black channel pieces to print in the current inventory. A one-material white shell still needs physical testing for light leakage; geometry and opacity are different properties.

Installation sequence

1. Use Q15 to test the actual LED strip at a complete corner. Check PCB width, component height, adhesive thickness, cut points, bend direction and connector/solder bulk. Do not assume a flat strip can bend edgewise around the rounded perimeter.

2. Feed the strip through the open service side and seat it against the defined perimeter carrier. The reference envelopes R04/R05 show reserved volume; they are not printed strips or proof of insertion feasibility.

3. Position a feed/corner joint where it can be accessed without removing a thread-bearing boss. Insulate solder joints and provide slack for service, without letting wires cross the optical belt.

4. Route head LED and tablet charging leads through B07's explicit internal duct. Route base LED leads within the base's service region, restrained away from the lower-cover screws and rear cartridge.

5. Use the rear conical cable funnel for the real incoming cable; it is a geometric guide, not a certified gland. Confirm strain relief with the actual cable and suitable real electrical components.

6. Close covers temporarily and inspect the belt in a dark room for unwanted emission through seams, vents and large white surfaces. Correct leakage with a compatible validated treatment or design change, not with an assumption based on the render.

Diffusion and electrical limits

The structural light belt is integrated with the shell; the tiny optical film O05 covers the decorative sensor detail, not the LED belt. White PETG transmission varies by pigment, wall thickness, layer orientation and finish. Compare Q08/Q09/Q10 transmission coupons before selecting brightness. The desired continuous line is a target; uniform corners and invisible cut joints are not established by CAD.

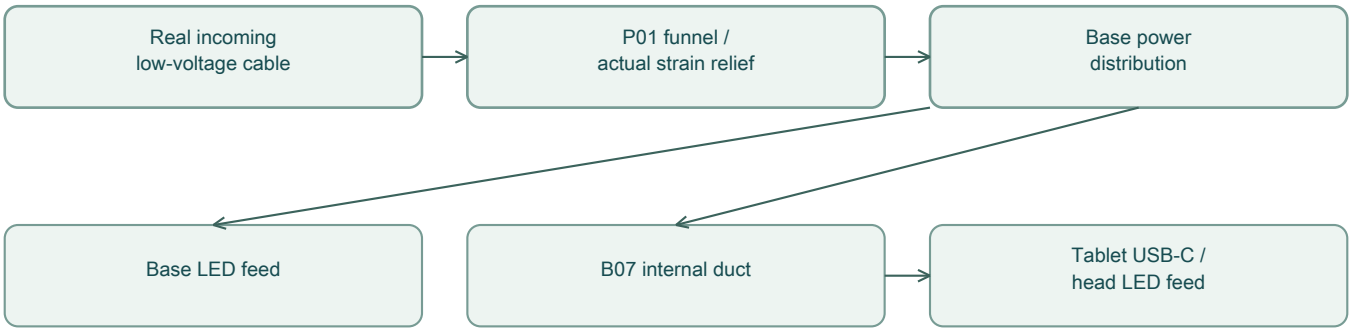
Do not infer strip voltage or power from the 8 mm width. Use the actual manufacturer's strip data. A printed plastic channel is not a conductive heat sink. If the strip requires a dissipating carrier, add a suitable real carrier and reassess the available space. Test the closed assembly at intended brightness and duration with temperature measurements at strip, printed support, power electronics and tablet. No universal safe wattage or charging capability is specified here.

Ventilation

The head and base include repeated vent patterns and fine concealment features. In slicer preview, confirm that the intended openings remain open and that the thin lattice is actually produced by the nozzle. These features are decorative/dust-obscuring geometry, not an IP-rated dust filter or quantified airflow system. Keep their paths free of infill and wiring; evaluate airflow and temperature with the actual electronics installed.

See the LED head, LED base, root reinforcement and cable section views. Orange LED envelopes are inspection references, not measured light distribution.

Cable topology and closure order



Intended interface relationships; the wiring view is a functional topology, not an electrical schematic.

This is a functional topology, not an electrical wiring schematic. Actual wire sizes, fusing, regulation, LED voltage, charging protocol and connectors depend on selected components. The optional ESP32 firmware is a separate low-voltage controller; it is not required for phone/tablet confirmation.

Before closing B03, pass the complete actual USB connector through B07 and the head joint. Add the head LED conductors alongside it and repeat the pull-through test. Leave the base LED feed within the base. Keep enough controlled slack to open O03 and disconnect the tablet, without a loose loop entering a vent or screw seat. Use actual suitable strain relief; the printed funnel alone is not retention.

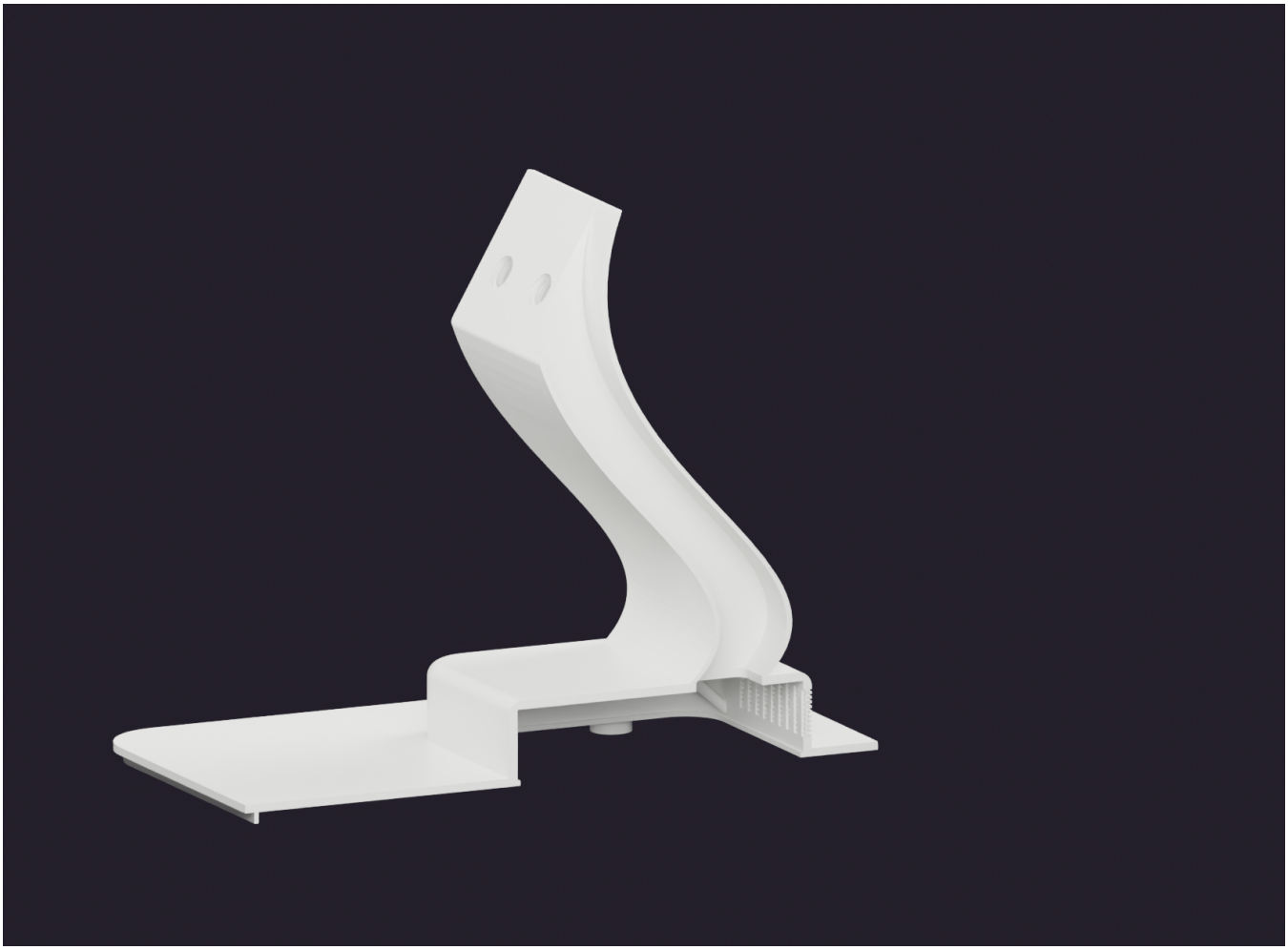


Figure 30. Opposite half-section of B07. This inspection-only subtraction is not part of the manufacturing files.

Figure 30 is a geometric inspection section. No visible external cable channel is added to the real stand. The section shows the modeled void; successful passage of a particular plug still requires its physical test.

Corner-feed planning checklist

- Identify the strip's permitted bend direction before cutting or attaching it.
- Reserve room for the real feed joint, insulation and cable bend, not only the 8 mm PCB.
- Keep the feed accessible from the service opening and clear of a thread seat.
- Trial-close the relevant cover with the unpowered strip and wiring in place.
- Verify electrical operation and intended luminous region before final closure.
- Run the closed-assembly optical/thermal checks in the acceptance plan.

Neither R04 nor R05 includes every possible solder joint or connector body. If the chosen strip needs a metal heat-spreading carrier or a larger bend, reassess the CAD clearance rather than forcing it into the reserved envelope.

9. Assembly Acceptance and Physical Test Plan



Figure 23. B07 and orange R02 only. The root braces belong to the stand, not the supply. Terminal/wiring/cooling clearance still needs the real component.

These are proposed build-qualification procedures, not completed test results. The verification record names the checks actually executed. Record part revision, material, print profile, actual electronics and ambient conditions with every result.

Gate A — Before large prints

Check	Method	Acceptance for the next stage
File scale	Import one known part and compare its bounds with the catalog	Millimetres and expected dimensions; no accidental unit scaling
Thread choice	Run the matching clearance coupons	Hand-start and repeated seating without damage
Tablet corner/opening	Fit actual tablet to Q07/Q13/Q14	No forced glass contact; dimensions recorded
USB connector	Compare actual molded body with Q12	Complete plug is represented; bends still to be checked in assembly
LED corner	Fit exact strip to Q15	Width, package height and corner-feed strategy are workable

Gate B — Unpowered complete dry assembly

1. Assemble B01/B07 and the head without live electronics. Confirm bearing faces seat before tightening.

2. Pass the actual charging connector through the complete internal route. Add the LED leads; repeat. Inspect both ends for pinch points with covers temporarily fitted.

3. Insert P01 and close the service assembly. Check that it remains recessed and cannot move outward under gentle handling. Do not insert a real plug into the decorative sockets.

4. Fit O01/O06/O05 and O03. With the tablet protected or absent, check capture from front and back. Reject loose films or faces that can fall out.

5. Fit the tablet without USB tension, close H05/H06, and confirm that stops bear on intended body regions. Check buttons, screen edges and front camera.

6. Repeat the documented upper-rear removal and reinstallation. Count any burr removal or adjustment; a screw should not be needed to pull a distorted cover into alignment.

7. Inspect all visible screw seating and the driver access to the spherical feet. Check whether the assembled base rocks on a flat desk.

Acceptance is successful operation of every relevant interface without damage, unintended preload or lost access. This procedure does not establish load capacity.

Gate C — Structural behavior

Support and restrain the prototype so the tablet cannot fall during testing. Begin with the intended dead load and a substitute mass if the real tablet is at risk. Observe stand-root deflection, the head joint, printed thread seats and the base after assembly. Evaluate touch forces and cable pull progressively, using measured loads selected for the intended use. Stop if a joint moves, a layer whitens/cracks or deformation grows.

Record force, application position, direction, duration and deflection; repeat after warm operation and after service cycles. Define the required load cases and allowed deformation before qualification. The illustrative moments in the design rationale are not passed test loads and no safe structural rating is assigned here.

Gate D — Powered closed-assembly behavior

Use the actual component documentation to establish supply, cable and LED limits. Start at a controlled low LED setting. Log strip/support, power-electronics and tablet-region temperatures in the closed assembly, including ambient conditions and time. Continue until the relevant temperatures stabilize under the intended use. Inspect for light leakage, charging interruptions and heat-induced movement.

A pass requires remaining within the applicable component/material limits and preserving mechanical fit and function. This repository does not specify a universal plastic-safe strip wattage or authorize an unverified mains installation. An open-bench ten-minute light test is not the closed-enclosure thermal test.

Gate E — Actual tablet / phone

Scenario	Expected behavior
One completed new enrollment	Exactly one named local profile appears
Three completed enrollments then app restart	All three records remain visible in Experiences
Open payment or validation QR	Request/waiting state; no immediate paid/valid result
Scan using Mobile with Bluetooth enabled	Response reaches the matching tablet session and result is accepted
Disable then re-enable tablet Bluetooth	Clear retry path; no stale attempt is accepted as a fresh confirmation
QR expiry followed by new request	New code is required; an old session must not complete the new one
Press Back after Mobile success	Home, rather than an unintended app exit
Internet unavailable, Bluetooth enabled	Local QR/BLE path remains available

Automated protocol tests support some state logic; they do not replace radio, permission, camera or Android lifecycle checks on the actual devices.

Test record template

Copy this into a build note for each gate:

- Date / operator / hardware revision / exact STL or native source.

- Printer / nozzle / filament / drying / layer height / orientation / walls / infill modifiers / supports.

- Actual tablet, phone, cable, LED and supply identifiers.

- Procedure and measured conditions.

- Observations, photographs and pass / fail / not tested.

- Corrective action and the revision that must be rechecked.

Do not replace a failed measurement with a render. Attach the evidence to the part IDs so future revisions can be compared.

10. Ten-Minute Offline Demo Runbook

This runbook is for VIA Terminal 1.0.5 on the tablet and VIA Mobile 1.0.5 on the phone. It keeps the presentation short while testing real local state transitions. The interface remains Romanian; English instructions below identify the corresponding controls.

Before the audience arrives

1. Install both official APKs from the release. Preserve existing local profiles by using a compatible official update rather than uninstalling.

2. Charge both devices. Enable Bluetooth on both, grant Camera and Nearby devices, and confirm that neither permission is blocked by Android settings. Older Android versions may also need location permission/settings for BLE discovery.

3. Choose steady frontal lighting. Keep the real front camera uncovered and clean; the black decorative sensor bar does not perform the recognition.

4. Open VIA Terminal and VIA Mobile. Rehearse one full QR transaction on the actual pair of devices. Internet is unnecessary, but Bluetooth remains necessary for the return confirmation.

5. For a quiet presentation, use the device's Do Not Disturb settings and a dedicated demo account where appropriate. This branch's immersive UI is not a guaranteed Device Owner kiosk or a universal notification blocker.

Suggested presentation sequence

Time budget	Demonstration	What the observer should see
0–2 minutes	Create a named profile: Profil nou → name → face → smile → palm	One saved profile for one completed enrollment, not a stream of new users
2–3 minutes	Open Experien□e	Every locally saved profile is listed; the count matches actual records
3–6 minutes	Open Plăte□te and choose the phone/QR route	Tablet displays a request and waits; success does not appear merely from tapping the option
6–8 minutes	Scan tablet QR using VIA Mobile	Phone sends the response locally over BLE; tablet shows the accepted result
8–9 minutes	Repeat with a validation service	Validation follows the scanned request and accepted response, not the service button tap
9–10 minutes	Press Back on Mobile, then show service access / interior figure	Phone returns Home; explain the hidden cable duct and removable upper-rear tablet restraint

The times are a presentation plan, not a performance measurement. With many participants, keep enrollment and demonstration transactions short; agree who operates the phone and who watches the tablet so cameras are not competing for the same view.

What each indication establishes

A face/smile/palm indication reflects local detector or landmark logic. It does not certify financial identity, liveness or resistance to impersonation. A signed QR/BLE result establishes that the demo accepted a bounded local protocol response. It does not transfer money or validate an externally issued travel entitlement. Demonstrate the interaction clearly without claiming stronger security than the implementation provides.

Recover in front of an audience

- Bluetooth was off: enable it fully, return to the tablet's Retry state and retry. If the phone has timed out too, retry there as well.

- QR expired: generate the new request and scan that one. Do not keep rescanning a screenshot of the old code.

- Phone is at the result screen: Back returns Home; start the next operation from there.

- Recognition is inconsistent: correct lighting and distance, clear the real front camera and retry one participant at a time. Do not present an unconfirmed state as success.

- A connection still fails: use the troubleshooting guide, rather than repeatedly enrolling duplicate users.

Session cleanup

Local data survives normal restarts. Decide before the event which demo profiles should remain; do not clear app data in the middle of a session without expecting profile loss. Do not publish participant biometric images, personal names or device backups with the CAD repository. The public build logs and CAD are separate from an audience's local data.

11. Troubleshooting by Symptom

Identify the responsible interface before sanding, scaling or tightening. Keep the part ID, actual file revision and a photograph in the build note.

Printing and mechanics

Symptom	Inspect	Corrective direction
Slicer shows infill in cable route	Import units, mesh interpretation, repair result, supports versus infill	Compare with duct-section figures; preserve the real CAD void
Head fits nominal box but not the printer preview	Brim/support extents and exclusion zones	Review diagonal placement and adhesion; do not crop a structural part to fit
Print needs inaccessible supports	Orientation and roofs inside carrier/duct	Reorient or revise the responsible feature before a full print; do not entomb support
Tablet opening seems too small	Actual dimensions, corner shape, first-layer/paint effects	Use gauges and local fit correction; avoid globally scaling all joints
H05 will not slide	H06 still engaged, guide burrs, paint, orientation	Remove H06 first, support tablet, inspect guide clearance
Tablet can move with covers closed	Stops, H05 seating, H06 pin engagement, wrong screw	Inspect complete restraint chain; do not pad the screen with a cable
Optical face or film falls forward	Correct stack, H01 shoulder, carrier seating, film thickness	Refit and test capture without the tablet; reject a loose assembly
P01 projects or moves outward	Recess seating, debris, service closure, mating contact	Measure actual interference/gap; revise cartridge/interface rather than leaving it proud
Stand rocks at root	Bearing face, burrs, layer damage, wrong screw family	Unload, inspect and correct fit; extra torque is not reinforcement
A screw cap is flush but joint is loose	Thread engagement, premature bottoming, stripped seat	Confirm receiving thread is doing work; replace damaged pieces
Fine vent lattice disappears	Minimum printable feature width and slicer line generation	Use actual preview and a local sample; a CAD line is not a printed strand
One-layer film lifts or has holes	First-layer height, cleanliness, flow and bed adhesion	Calibrate the 0.20 mm single layer; keep the outline undamaged

LED and charging

Symptom	Inspect	Corrective direction
Strip cannot turn a corner	Bend direction/radius, PCB construction, components	Follow actual strip data; plan accessible allowed cut/feed joints
Feed joint collides with a screw	Corner feed placement and service slack	Route around the bearing/thread region before closure
Whole white shell glows	Filament opacity, wall thickness, seams and finish	Test transmission coupons and a compatible masking/design solution
Belt is uneven	Strip-to-wall geometry, corners and joints	Test exact parts/strip; renders are not a luminance measurement
Tablet does not charge reliably	Supply voltage/protocol, cable, plug seating and strain	Verify the real charging chain outside the enclosure first
Temperatures keep rising	Actual strip power, carrier requirements, closed airflow	Stop the test and review the thermal design/component documentation

Tablet and phone

Symptom	Inspect	Corrective direction
New profile is not in Experiences	Enrollment completion and local save	Finish the full named flow; compare actual list after restart
Phone scan does not complete tablet	Bluetooth, Nearby devices, QR expiry, matching session	Enable both adapters, grant permissions, refresh/scan the current code
Retry after Bluetooth enable still stalls	Adapter fully on, phone timeout, stale session	Retry on both sides as needed; generate a fresh QR after expiry
Phone Back exits unexpectedly	Installed app version/signature	Use official 1.0.5 and retest from its result screen
Official APK will not update	Existing package ID, signing key, Android install error	Do not mix CI and official signatures; protect local data before uninstalling
Face/palm indications are weak	Real camera obstruction, lighting, distance, permissions	Correct the camera setup; decorative lenses are not detectors
Internet is absent	BLE and local permissions, rather than Wi-Fi	The app requires no server; keep Bluetooth enabled for confirmation
Android notifications appear	Device-level presentation configuration	Use OS DND/settings; immersive UI is not a proven Device Owner kiosk

See demo runbook and physical acceptance plan for the expected successful sequence.

12. Requirements-to-Evidence Matrix

This matrix condenses the design decisions and corrections that shaped VIA. “Present in CAD” and “physically validated” are deliberately different statuses. Hardware remains R14.1; the expanded figures explain the delivered geometry rather than inventing a further redesign.

Requirement	Delivered mechanism / artifact	Evidence and remaining check
Fits OnePlus Pad OPD2203	Nominal body envelope, H01 pocket, Q07/Q11/Q13/Q14	Native CAD and gauges; actual tablet fit/camera still required
Original front glass; no obvious tablet clamps	Structural surround and rear stops	Product/front views; inspect screen overlap and camera on device
Tablet removable through upper rear	H05 sliding bar and H06 integrated locking pins	Historical service sweep and figures 20/21/29; real removal test
Continuous one-piece S stand	B07 integrates stand and upper power-volume cover	One required B07 STL; layer/load/creep tests pending
Reinforcement belongs to the stand	Integrated root braces	Figures 09/23; structural load qualification pending
Cables hidden through the stand	Explicit modeled duct	Figures 10/30 and Q12; actual plug plus LED leads must pass
Monoblock outer base	B01	Required STL; trial seam and cartridge seating
Bottom service access	B03 / V41–V44	Figure 24 and service map; actual service access check
No added visible top screw holes	Underside-access base assembly	Assembled views and CAD; verify actual screw seating
Integrated head/base LED channel structures	H01 and B01	Figures 07/08; no L1/L2 items in print inventory
8 mm LED plus allowance	8.5 mm nominal reserved width	Q15; actual bend, solder joint and thermal needs
Light only at intended belt	Integrated rim/shield geometry, about 3 mm belt	Optical transmission intent; leakage/brightness test pending
More realistic black dummy ports	One P01 cartridge, recessed	Figures 04/25/26; visual and retention checks
No projecting port plate	Base recess and cartridge position	CAD/product views; printed tolerance and closure check
Complex dummy sensor appearance	Separate black O06 face inside O01	Figures 11/27/33; printed detail depends on nozzle/settings
Captive optical face / film	H01 shoulder, carrier relationship and O03 service access	Figure 27; verify capture in both directions after printing
One-layer optical film	O05 0.20 mm, original outline preserved	Native amendment and STL audit; adhesion/retention trial
No tiny loose lock pin	H06 pins integral to cover	R14 source and service map; printed pin engagement test
Printable fasteners	Coarse D12/D8 screw families	Thread meshes/coupons; no ISO metric or torque claim
Hidden part identification	Stable IDs and interior marks	Catalog and native labels; ensure marks survive slicing/finish
Minimal-support strategy	Open O01, flat-backed O06, guided bar without snap arms	Supplied orientation; actual slicer support review pending
All structural/dummy parts printable	Color/quantity inventory	32 audited STL files; electronics remain real components
No plate arrangement	Individual STL files by color	Print folder and release archive

Requirement	Delivered mechanism / artifact	Evidence and remaining check
Entire demo local	No INTERNET permission; local profiles and QR/BLE	Build/manifest checks; actual device interoperability test
No instant payment/validation	Bounded signed request/response flow	Protocol tests and demo runbook
Multiple named profiles	Atomic save, complete local list	Persistence/UI tests; restart check on tablet
Correct retry / Mobile Back	Adapter retry handling; Home navigation	Regression tests; lifecycle/radio checks on devices
Inspect a normal assembled model	Delivered English-labeled Blender scene	Opens assembled; sectioned figures are separate illustrations

Source artifacts: CAD provenance, verification, parts and quantities, software. There is no claim of a certified payment system, dust filter, thermal rating or secure biometric identity.